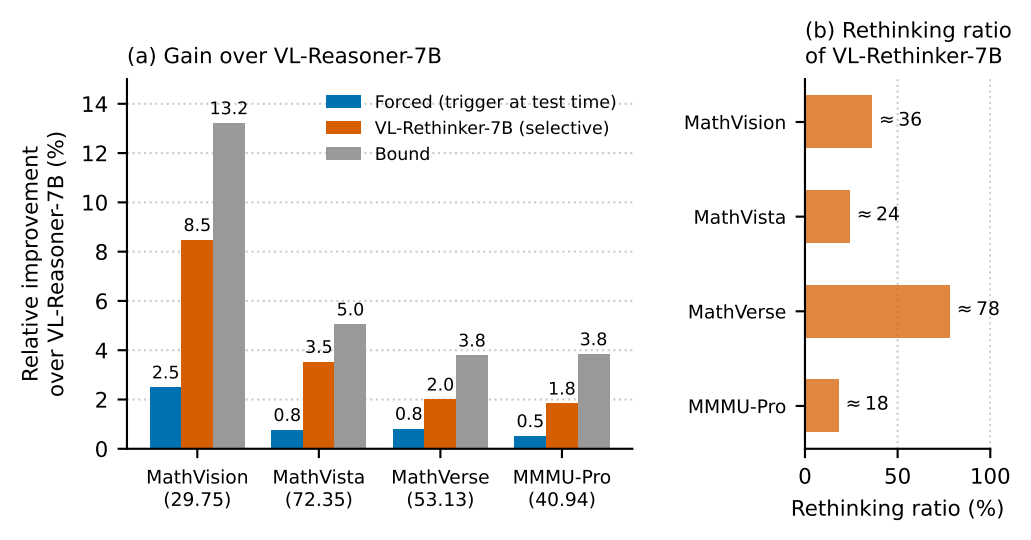
(a) Gain over VL-Reasoner-7B
Relative improvement over VL-Reasoner-7B (%)
Forced (trigger at test time)
VL-Rethinker-7B (selective)
Bound
0
2
4
6
8
10
12
14
2.5
8.5
13.2
0.8
3.5
5.0
0.8
2.0
3.8
0.5
1.8
3.8
MathVision (29.75)
MathVista (72.35)
MathVerse (53.13)
MMMU-Pro (40.94)
(b) Rethinking ratio of VL-Rethinker-7B
MathVision
MathVista
MathVerse
MMMU-Pro
≈ 36
≈ 24
≈ 78
≈ 18
0
50
100
Rethinking ratio (%)